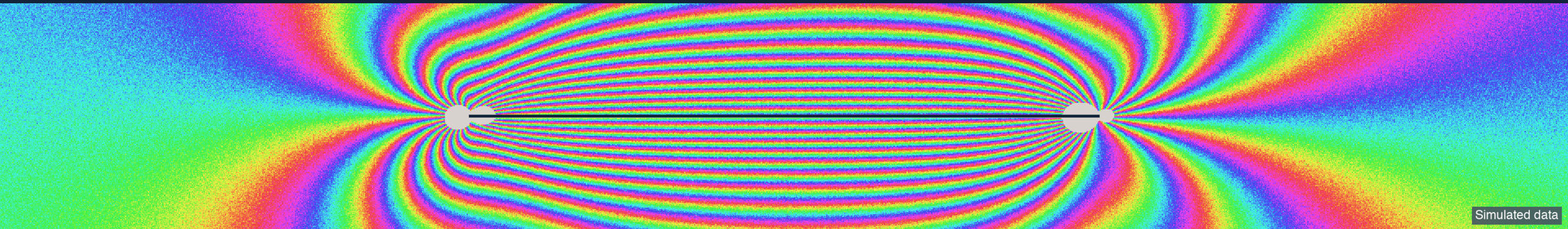
ENVIRONMENTAL GEOHAZARDS ENGINEERING · LECTURE

Earthquake Monitoring with InSAR

Measuring how the ground moves before, during and after earthquakes

AKSHAY RAJ MANOCHA



GOAL OF THIS LECTURE

Use satellite radar to map how the ground moved in an earthquake, find the fault that slipped, and measure the strain building up for the next one.

How to read coseismic fringes, how fault models are fitted to them, and how engineers use the results.

THIS LECTURE

Outline

01 Earthquakes and the earthquake cycle
Faults, elastic rebound, magnitude, and what InSAR adds

02 Coseismic interferograms
Reading the fringes of strike-slip, thrust and normal earthquakes

03 From fringes to the fault
Fault models, inversion and magnitude from geodesy

04 Before and after the earthquake
Strain build-up, seismic hazard, afterslip and triggered hazards

05 Case studies
Gorkha 2015, Türkiye 2023, Myanmar 2025 and rapid-response services

06 Engineering practice
Fault rupture, damage mapping, limits and exercises

01

PART 1

Earthquakes and the earthquake cycle

Faults, elastic rebound, magnitude, and what InSAR adds

EARTHQUAKES

Three ways a fault can move

Strike-slip

The two sides slide past each other horizontally. Examples: San Andreas (USA), North Anatolian (Türkiye), Sagaing (Myanmar).

InSAR sees: horizontal motion, strongly dependent on the fault's direction

Thrust (reverse)

One side is pushed up and over the other, under compression. Example: the Himalayan front, where India pushes into Asia.

InSAR sees: uplift over the upper (hanging-wall) block

Normal

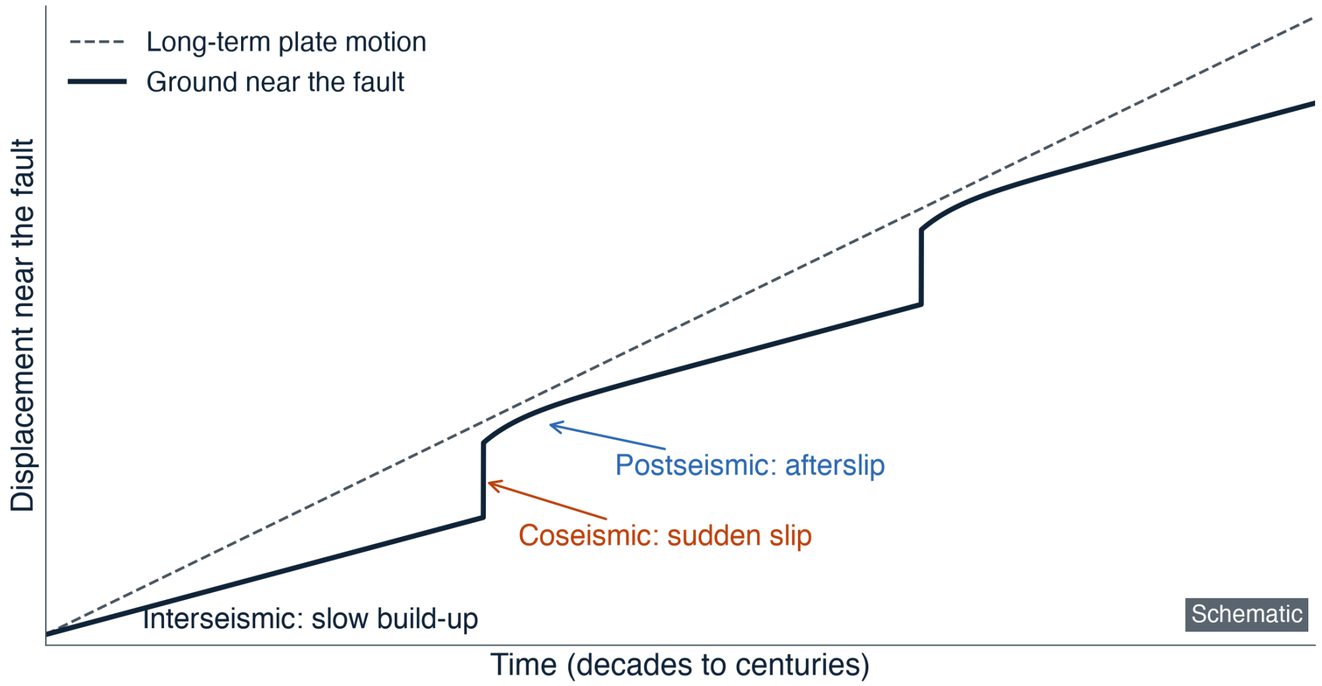
One side slides down, under extension. Examples: the East African Rift, the central Apennines (Italy).

InSAR sees: subsidence over the hanging-wall block

Real earthquakes often mix two: oblique slip, or several faults breaking together.

EARTHQUAKES

The earthquake cycle



Schematic: the ground near a locked fault lags behind the plates, then catches up.

Interseismic: decades to centuries of slow bending, a few mm per year.

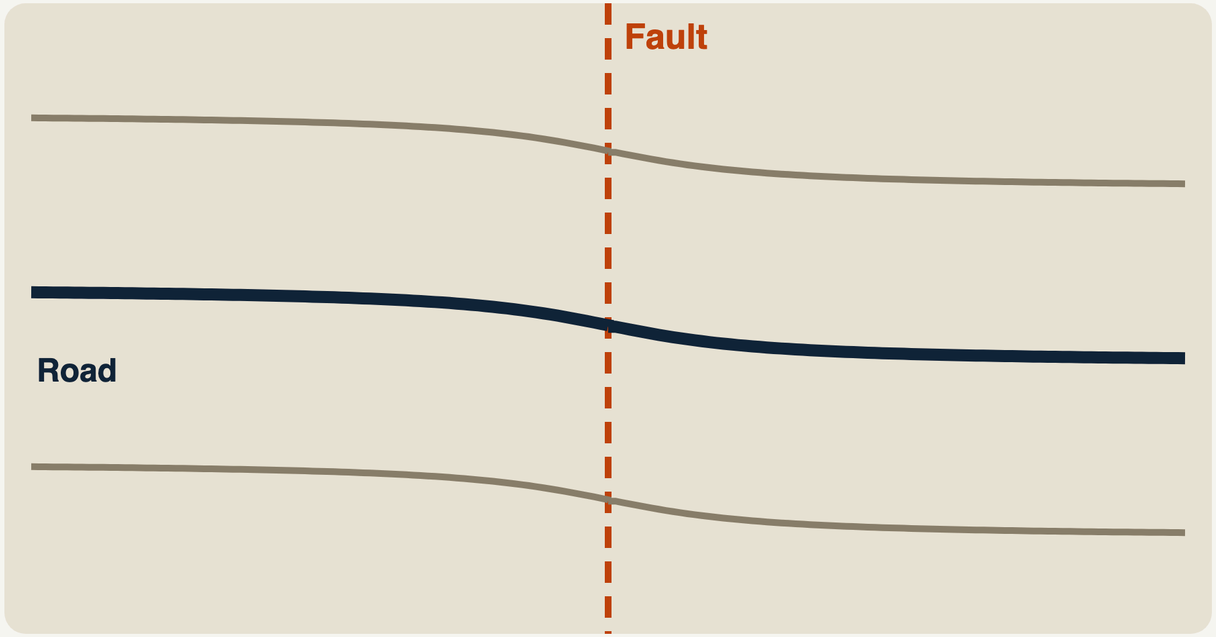
Coseismic: seconds of slip, centimetres to metres.

Postseismic: weeks to years of afterslip and relaxation.

InSAR measures all three, with different methods.

EARTHQUAKES

Elastic rebound: bend, snap, repeat



Map view, motion exaggerated

Locked fault: the ground bends

Interseismic | Coseismic | Postseismic

What InSAR measures

▶ **Slow bending: mm per year, over decades**

The jump: centimetres to metres in one day

Afterslip and relaxation: weeks to years

Elastic rebound (Reid, 1910)

EARTHQUAKES

How big? Moment magnitude

$$M_0 = \mu \times A \times D$$

Seismic moment: rigidity μ (about 30 GPa) × fault area A × average slip D.

$$M_W = \tfrac{2}{3}\,(\log_{10} M_0 - 9.1)$$

Each step of 1 in Mw is about 32 times more moment.

Length	Width	Slip	Moment	Mw
10 km	8 km	0.3 m	7.2×10^{17} N·m	5.8
40 km	15 km	1.5 m	2.7×10^{19} N·m	6.9
150 km	20 km	4.0 m	3.6×10^{20} N·m	7.6
500 km	20 km	4.0 m	1.2×10^{21} N·m	8.0

EARTHQUAKES

What InSAR adds to seismology

1992
Landers, California: the first earthquake mapped by InSAR (ERS-1)

250 km
Width of one Sentinel-1 image: a whole rupture at once

Days
Wait for the first image after an earthquake

~1 cm
Precision of one coseismic interferogram

Where the fault is

Maps the rupture even where it does not break the surface, as in the 2015 Gorkha earthquake.

How it slipped

Thousands of measurements, not a few GNSS stations: enough to map slip on the fault.

But not an alarm

Images are days apart: no early warning. Seismic networks do that job.

02

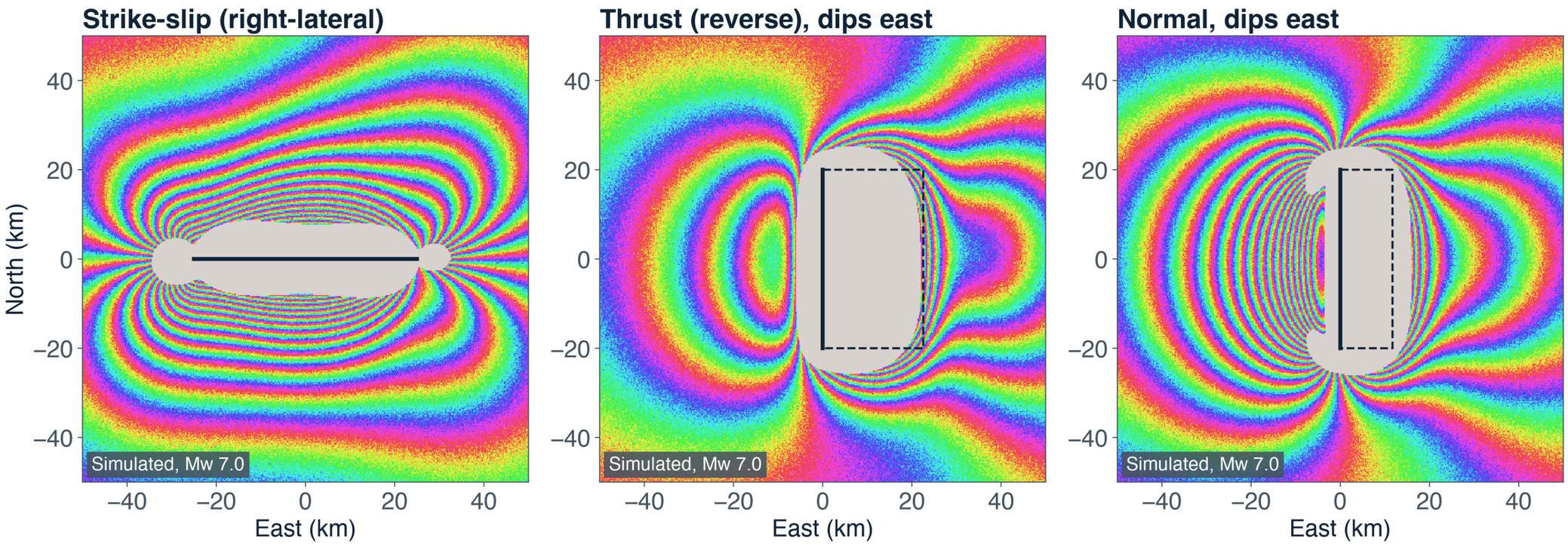
PART 2

Coseismic interferograms

Reading the fringes of strike-slip, thrust and normal earthquakes

COSEISMIC

What each kind of earthquake looks like

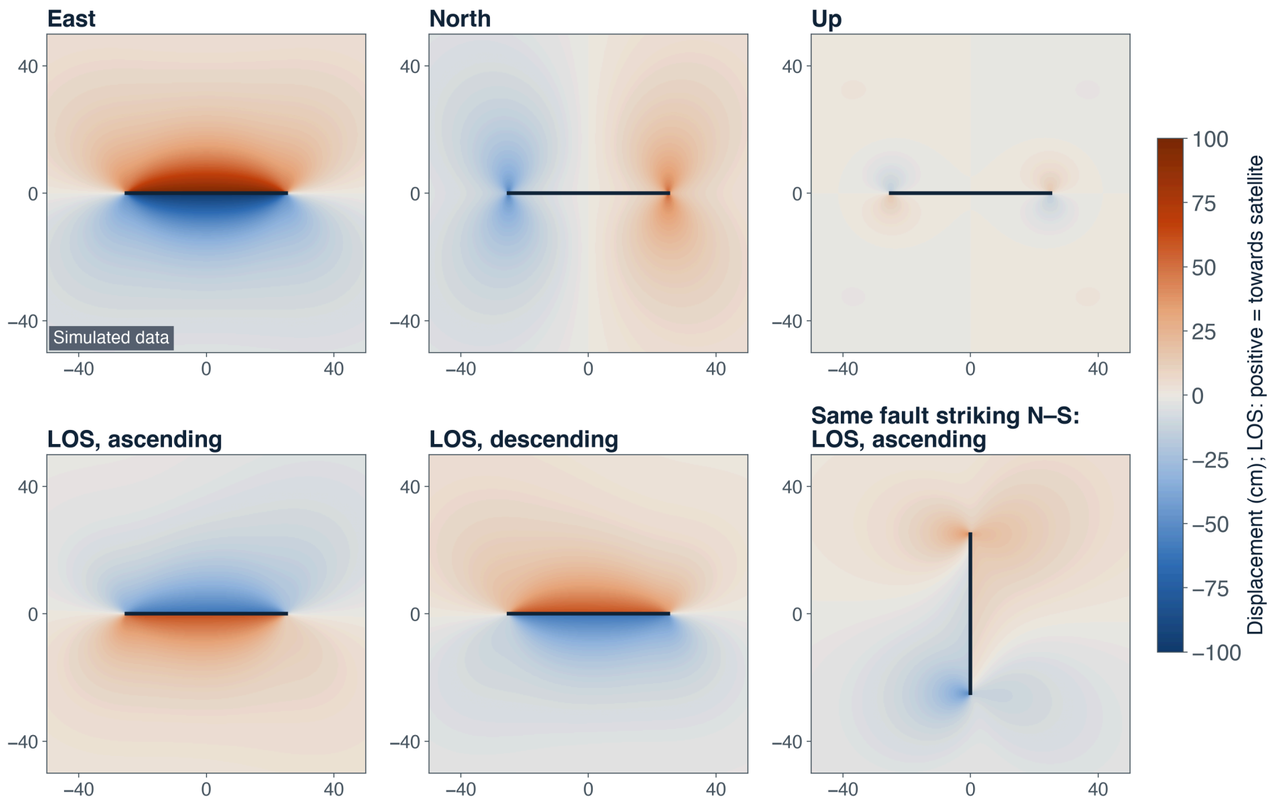


Sentinel-1 ascending pass. One colour cycle = 2.8 cm of motion along the line of sight. Grey: fringes too dense to resolve.

Simulated (Okada model), Mw 7.0 each. Peak motion along the line of sight: strike-slip ±63 cm, thrust uplift up to 81 cm, normal subsidence up to 118 cm.

COSEISMIC

One earthquake, many views



Simulated: east–west right-lateral fault, 2 m slip.

The line of sight mixes east, north and up into one number.

Ascending and descending see the same earthquake with opposite colours: they look from opposite sides.

Turn the fault north–south: the motion along it is almost invisible. Only the fault ends still show.

COSEISMIC

Reading a coseismic interferogram

01 **Find the fault:** fringes crowd together and coherence is lost along the rupture.

02 **Count the fringes** from the far field towards the fault: each one is 2.8 cm (Sentinel-1).

03 **Find the sign:** unwrap, or read the colour order, to tell towards from away.

04 **Read the pattern:** lobes along the fault (strike-slip), one big lobe over one side (thrust or normal).

05 **Compare both passes,** then fit a fault model.

COSEISMIC

Near the fault: too many fringes

Why the signal is lost

Surface rupture, collapsed buildings and landslides destroy coherence; metres of motion in a few kilometres give fringes too dense to count.

Offset tracking

Match image patches before and after: shifts in range and along the track, any size, decimetre precision.

L-band helps

One NISAR or ALOS-2 fringe is about 12 cm instead of 2.8 cm, and forests stay coherent.

Other tools

Burst-overlap interferometry for motion along the track; optical image correlation (Sentinel-2) for horizontal offsets.

03

PART 3

From fringes to the fault

Fault models, inversion and magnitude from geodesy

FAULT MODELS

A fault model predicts the fringes

The standard model is a rectangular fault in an elastic half-space (Okada, 1985). Give it:

- **Location and length**
- **Strike (direction)**
- **Dip (steepness)**
- **Top and bottom depth**
- **Slip amount**
- **Rake (slip direction)**

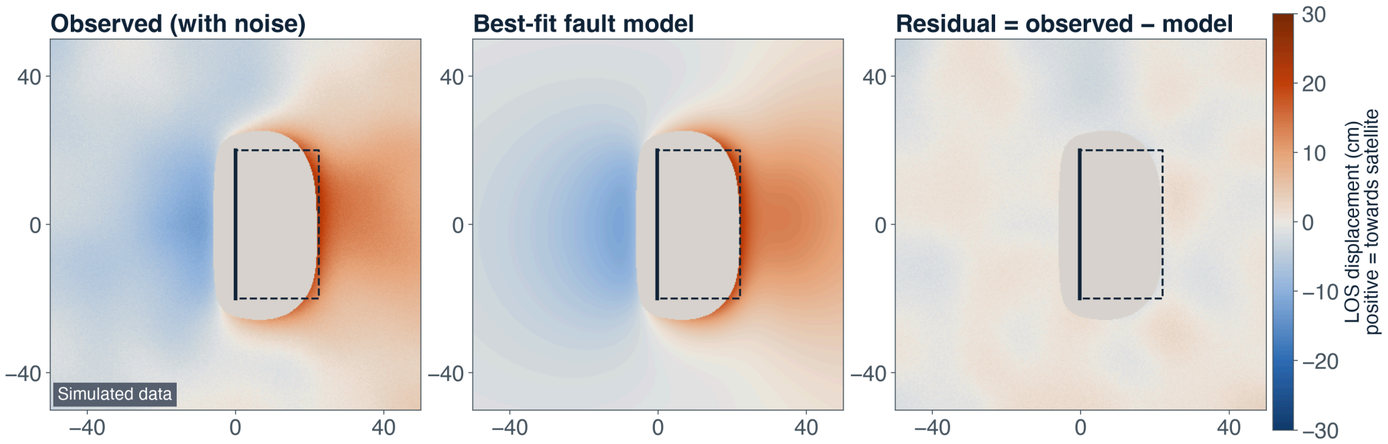
It returns the east, north and up displacement at every point, which is then projected onto the line of sight.

Used in this lecture

Every simulated interferogram here comes from Okada’s equations, checked against the test values published in his 1985 paper.

FAULT MODELS

Inversion: from fringes back to the fault



Simulated data (Okada fault model): data, best-fit model, and what is left (about 1 cm, mostly atmosphere).

	True	Fitted
Slip	1.5 m	1.7 m
Dip	30°	27°
Depth range	2–15 km	1.5–13 km
Mw	7.05	7.07

> The fault is adjusted until its predicted fringes match the data (least squares).

> Slip and depth trade off, but the magnitude comes out right.

FAULT MODELS

What fault models tell us

Which fault moved

Not always the obvious one: InSAR has found ruptures on unmapped faults and on several faults at once.

Magnitude from geodesy

An independent check on the seismic magnitude. A bigger geodetic value can mean afterslip inside the image pair.

Where slip stopped

Unruptured parts of a fault next to the rupture are where stress has increased.

Depth of slip

Shallow slip means stronger shaking and surface offsets; slip that stops at depth leaves a shallow hazard.

04

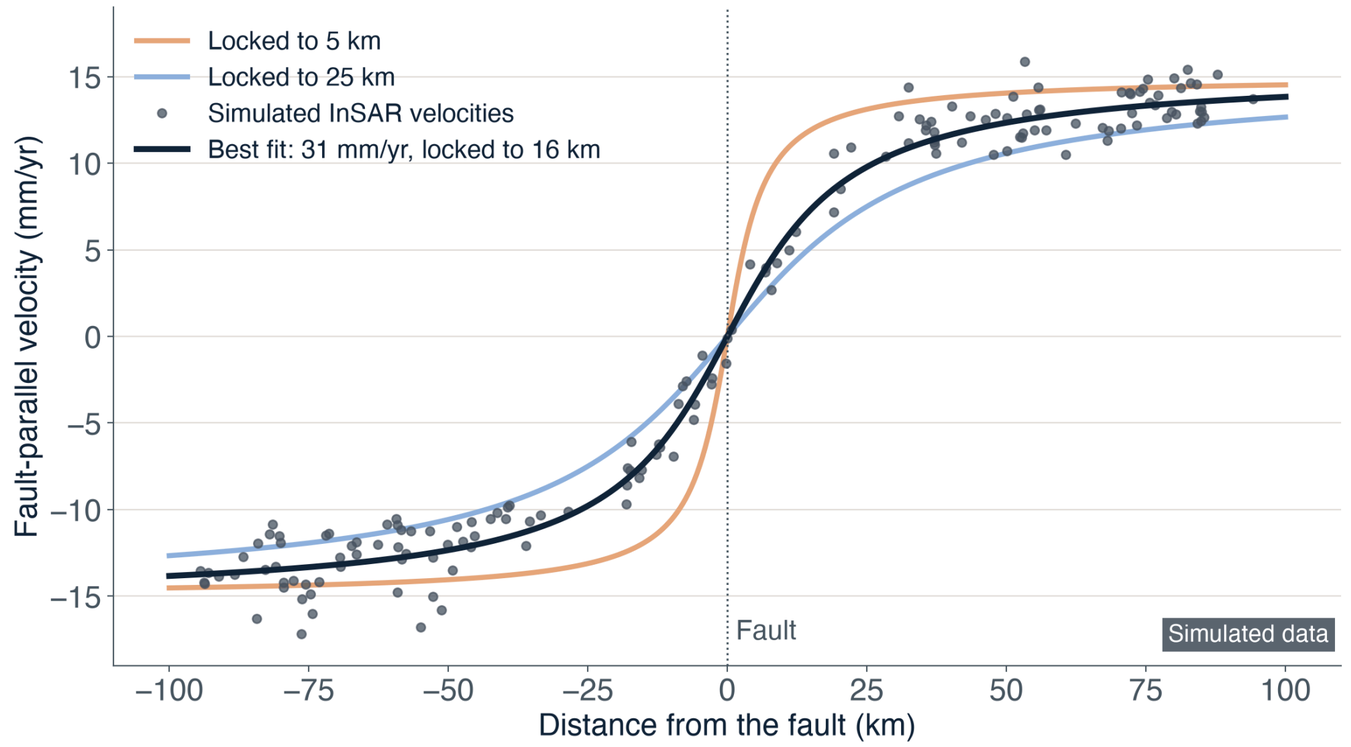
PART 4

Before and after the earthquake

Strain build-up, seismic hazard, afterslip and triggered hazards

INTERSEISMIC

Between earthquakes: measuring the strain



Simulated data: true values 30 mm/yr, locked to 15 km.

Across a locked fault the velocity follows an S-shaped curve: **v = (s/π) arctan(x/D)**.

Its height gives the slip rate s; its width gives the locking depth D. Fit: 31 mm/yr, 16 km.

Needs years of images, mm/yr precision, and GNSS to tie it to the plates.

INTERSEISMIC

From strain to seismic hazard

≈ 2 cm/yr
India converging with Asia across the Himalaya

10 m
Slip deficit after 500 years if the fault stays locked (500 × 2 cm)

A locked fault stores the plate motion as a slip deficit: the slip of a future earthquake.

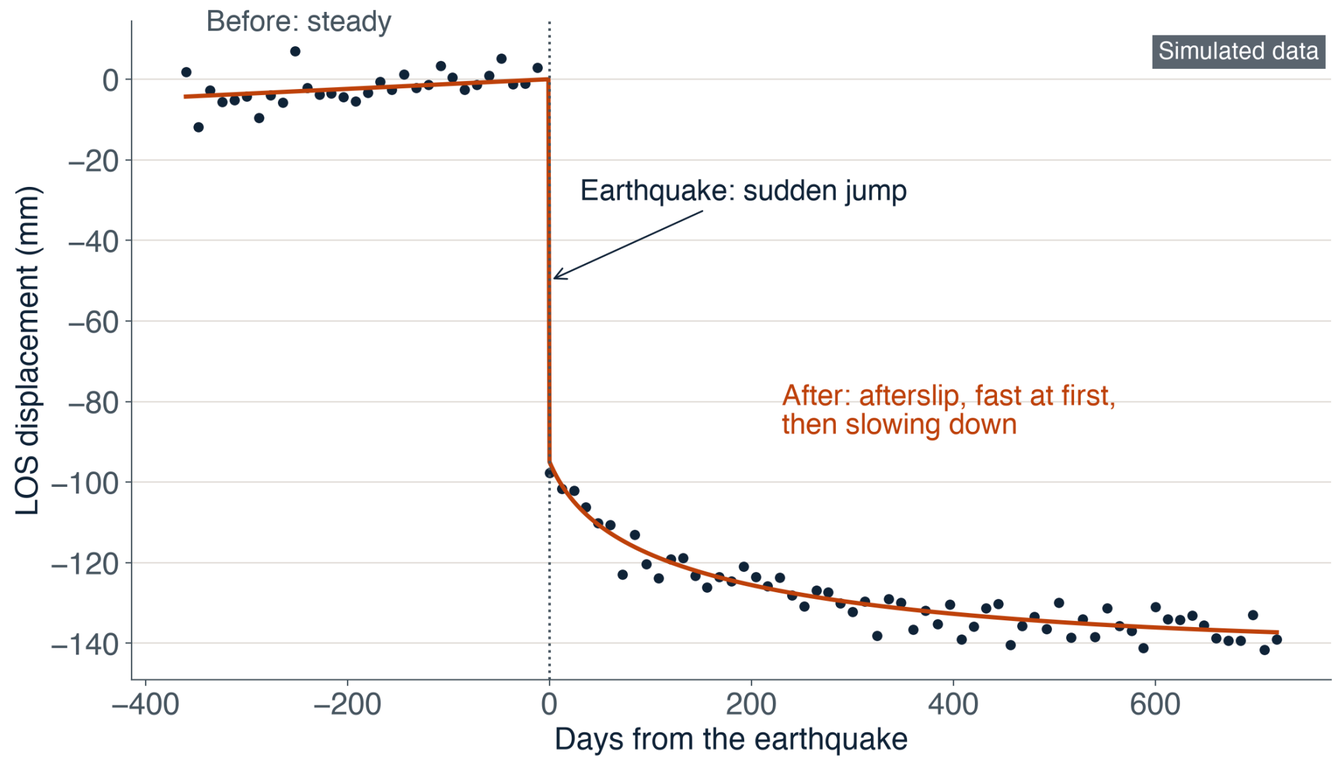
Strain-rate maps from InSAR and GNSS show which faults are loading, and how fast.

Some faults creep steadily instead, releasing strain without large earthquakes.

Hazard models combine these rates with earthquake history and geology.

POSTSEISMIC

After the earthquake, the ground keeps moving



Simulated: Sentinel-1 every 12 days.

Afterslip: the fault keeps slipping slowly, fast at first, then decaying over months.

Relaxation: the hot lower crust and mantle flow, for years to decades.

Poroelastic: groundwater flows back after the stress change, for weeks to months.

Here afterslip adds about 41 mm in the first year, about 43% of the jump.

POSTSEISMIC

Earthquakes trigger other hazards

Landslides

Strong shaking in mountains triggers thousands of landslides. Coherence loss and offsets map them; time series follow slopes left unstable.

Liquefaction

Saturated sand loses strength: ground sinks and spreads near rivers and coasts. InSAR maps the settlement.

Damage

Collapsed buildings break coherence. Damage proxy maps turn that into a rapid map of where to look (Part 6).

05

PART 5

Case studies

Gorkha 2015, Türkiye 2023, Myanmar 2025 and rapid-response services

CASE STUDY

Gorkha, Nepal, 2015

Mw 7.8
25 April 2015, on the Main Himalayan Thrust

≈ 1.5 m
Largest ground displacement, about 20 km north-east of Kathmandu

0 km
Surface rupture: the fault did not break the surface

3
Radar satellites used: ALOS-2, Sentinel-1, RADARSAT-2

InSAR mapped a blind thrust: slip stopped beneath Kathmandu, which was lifted up.

The shallow part of the fault to the south did not slip, so its hazard remains.

L-band ALOS-2 kept coherence on the forested Himalayan slopes.

Sources: Lindsey et al. (2015), Geophysical Research Letters; ALOS-2 ScanSAR study, Earth, Planets and Space (2015).

CASE STUDY

Kahramanmaraş, Türkiye, 2023

Mw 7.8 + 7.5
Two earthquakes about nine hours apart, 6 February 2023

310 + 140
Kilometres of rupture, mapped with Sentinel-1 offsets

7–9 m
Largest offsets along the fault strands

> 50,000
People killed in Türkiye and Syria

Near the faults the fringes were far too dense: offset tracking traced the ruptures.

Burst-overlap interferometry measured motion along the satellite track.

NASA–JPL ARIA damage proxy maps from Sentinel-1 coherence guided the response.

Sources: USGS fault rupture mapping from satellite data (2023); Geophysical Journal International 236, 1068 (2024); NASA JPL ARIA.

CASE STUDY

Mandalay, Myanmar, 2025

Mw 7.7
28 March 2025, on the Sagaing Fault

≈ 500 km
Rupture length, one of the longest on land

3–5 m
Offsets along the fault from offset tracking

N–S
Fault direction: the hardest case for InSAR

The fault runs north–south, so most of the motion was along the satellite track.

Offset tracking and optical correlation measured it; interferometry added the smaller vertical and east–west parts.

The rupture ran faster than shear waves (supershear) for hundreds of kilometres.

Sources: Communications Earth & Environment (2025), ‘Sustained supershear rupture during the 2025 Mandalay, Myanmar earthquake’; PNAS (2025).

CASE STUDY

Rapid response from space

COMET-LiCSAR

Automatic Sentinel-1 interferograms, triggered by the USGS earthquake catalogue: about an hour after a new image arrives.

NASA–JPL ARIA

Damage proxy maps from coherence loss, released to responders within days.

USGS

Magnitude, finite-fault models and ShakeMap from seismology, updated with satellite results.

Space agencies

Satellites are tasked to image disaster areas, and NISAR adds open L-band data.

06

PART 6

Engineering practice

Fault rupture, damage mapping, limits and exercises

ENGINEERING

Where engineers use earthquake InSAR

Fault setback zones

Map active fault traces and keep buildings off them.

Lifelines across faults

Pipelines, canals, railways and roads designed for the expected offset.

Dams and bridges

Check for deformation after an earthquake, before reopening.

Damage mapping

Damage proxy maps send inspectors to the worst areas first.

Landslides, liquefaction

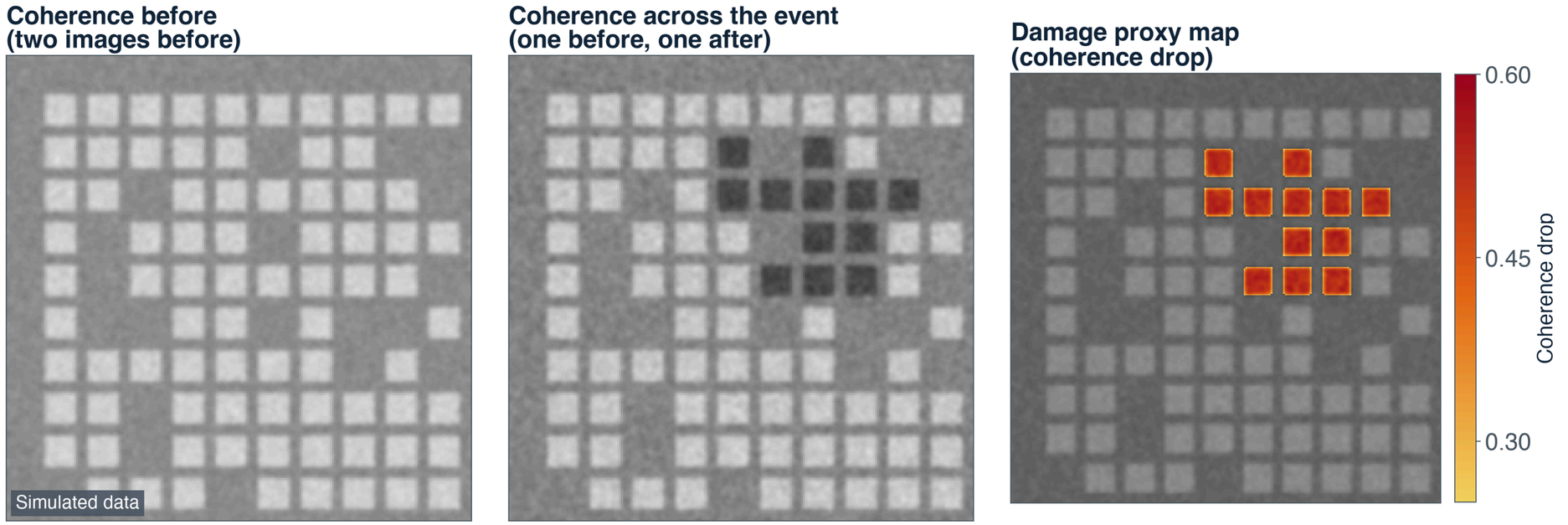
Find ground failures triggered by the shaking.

Hazard models

Slip rates and strain rates feed seismic hazard maps and design codes.

ENGINEERING

Damage proxy maps from coherence



Simulated town. Damaged blocks lose coherence across the earthquake; the drop is the proxy for damage.

Method: compare coherence of a pre-event pair with a pair spanning the event.

Caution: crops, construction and vegetation also lose coherence. Check with optical images.

ENGINEERING

Limits and good practice

Limits

- One direction only; north–south motion nearly invisible
- Dense fringes and lost coherence near the rupture
- Atmosphere comparable to mm/yr strain signals
- Images days apart: no early warning

Good practice

- Use both passes plus offsets for big earthquakes
- Tie long time series to GNSS
- Correct the atmosphere; use long time series
- Combine with seismology and field mapping

EXERCISES

Try these

01 A coseismic interferogram shows 18 fringes between the far field and the edge of the decorrelated zone. How much line-of-sight displacement is that?

02 A rupture is 30 km long and 12 km wide with an average slip of 1 m (μ = 30 GPa). What is its moment magnitude?

03 Across a locked fault slipping 30 mm/yr, the velocity is 7.5 mm/yr at 15 km from the fault. What is the locking depth?

04 Next to a strike-slip fault the ground moves 1.5 m. How much does an ascending pass see if the motion is to the north? To the east?

EXERCISES

Answers

01 18 × 2.8 cm ≈ **50 cm** along the line of sight (the sign needs unwrapping or the colour order).

02 Moment = 3×10^{10} × 30,000 × 12,000 × 1 = 1.1×10^{19} N·m, so Mw = ⅔ (19.03 − 9.1) ≈ **6.6**.

03 7.5 = (30/π) arctan(15/D), so arctan(15/D) = π/4 and **D = 15 km**: the velocity reaches a quarter of the slip rate at x = D.

04 North: 1.5 × 0.11 ≈ **16 cm**. East: 1.5 × 0.62 ≈ **93 cm**. Both away from the satellite; use offsets for the north–south case.

RECAP

Key points

01 Earthquakes release, in seconds, strain that built up over centuries.

02 Coseismic fringes reveal the fault type, the moving block and the amount of motion.

03 Fault models fitted to InSAR give fault geometry, slip and magnitude.

04 Between earthquakes, InSAR measures slip rates and locking depths for hazard maps.

05 Engineers use it for fault zoning, lifelines, damage mapping and post-event checks.

FURTHER READING

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Thank you. Questions?

Figures and the animation were made for this lecture. Interferograms, the inversion, the time series and the damage map are simulated, and labelled as such; coseismic fields use the Okada (1985) model with Sentinel-1 wavelength and geometry. Case-study facts: Lindsey et al. (2015) and the ALOS-2 study in Earth, Planets and Space (Gorkha); USGS, Geophysical Journal International and NASA JPL ARIA (Türkiye); Communications Earth & Environment and PNAS (Myanmar); Lazecký et al. (2020) for COMET-LiCSAR. Sentinel-1 data: EU Copernicus programme / ESA. Full references on the previous slide.